

Distance Squared between Two Points

DESCRIPTION

In a 2D Cartesian plane, the squared Euclidean distance between two points (x_1, y_1) and (x_2, y_2) is defined as:

$$D^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

Given the integer coordinates of two points, compute D^2 .

INPUT FORMAT

The input consists of four integers x_1, y_1, x_2, y_2 , separated by spaces.

OUTPUT FORMAT

Print the value of the squared distance D^2 , followed by a newline.

SAMPLE

Sample Input	Sample Output
1 2 4 6	25